

Shikang Wen

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Hong Kong

RESEARCH INTERESTS

- Carbon neutral building

EDUCATION EXPERIENCE

Bachelor of Architecture, Shanghai Jiao Tong University
PhD candidate, The Hong Kong Polytechnic University

2018.09 – 2023.06
2023.09 – now

ACADEMIC PUBLICATIONS

1. **S. Wen**, X. Hu, G. Hua, P. Xue, and D. Lai, Comparing the Performance of Four Shading Strategies Based on a Multi-Objective Genetic Algorithm: A Case Study in a University Library, 2022. (**Journal of Building Engineering, IF: 7.4**)
2. **S. Wen**, J. Xue, G. Hua, and D. Lai, Reconstructing fisheye luminance maps with a two-step network from a single low dynamic range image, 2024. (**Automation in construction, IF: 11.5**)
3. **S. Wen**, R. You, and Q. Chen, Implementing building retrofitting strategies to halve campus office building carbon emissions by 2035: A case study in Hong Kong with techno-economic analysis, 2025. (**Energy and Buildings, IF: 7.1**)
4. **S. Wen**, B. Shen, Y. Yao, and Q. Chen, Energy performance and life-cycle analyses of advanced terminal devices and heat pumps for a medium-sized office building in different climate zones in China, 2025. (**Energy and Buildings, IF: 7.1**)
5. **S. Wen**, R. You, and Q. Chen, Energy-exergy performance and indoor comfort of a radiant ceiling cooling system with air-to-air sensible heat exchanger in hot and humid climate, 2025. (**Building and Environment, IF: 7.6**)
6. **S. Wen**, R. You, and Q. Chen, A novel automated energy-efficient building design: A case study in a campus office building, 2025. (**Submitted to Energy**)
7. **S. Wen** and Q. Chen, A novel building design method through 2D-to-3D reconstruction: Reconstructing from 2D images to 3D building energy modeling inputs, 2025. (**Applied Energy, Under Review**)

RESEARCH EXPERIENCE

1. **Optimization of Building daylight space** Mar 2021 - Jun 2022
Under the guidance of Prof. Lai, I developed a framework to compare various shading strategies, incorporating a performance curve generated from a multi-objective genetic algorithm. This experience sharpened my research, simulation, optimization, and academic writing skills.
Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.
2. **Real-time glare detection** Jul 2022 - Mar 2023
An academic project, supervised by Prof. Lai and Prof. Xue, I developed a deep learning approach to reconstruct luminance maps from single-exposure images, honing my skills in generative adversarial networks (GANs) and high dynamic range (HDR) photo analysis.
Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.
3. **Building retrofits in high-density campus** Sep 2023 - May 2024
Under the supervision of Prof. Chen and Prof. You during my PhD project, I developed expertise in building retrofitting strategies, research methodology, building energy simulation tools, and academic writing.
Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.
4. **Advanced radiant ceiling cooling system** May 2024 - Jan 2025
Under the supervision of Prof. Chen during my PhD project, I mastered the analysis of radiant cooling systems and specialized in problem-solving research methodologies.
Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.
5. **Automated building form design** Feb 2025 - Jun 2025
During my doctoral studies under the supervision of Prof. Chen and Prof. You, I mastered the automated building form generation algorithm and specialized in developing automated building form generation to the EnergyPlus model.
Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.
6. **Automated 2D image to 3D EnergyPlus model** May 2025 - Nov 2025

During my doctoral studies under Prof. Chen’s supervision, I specialized in developing automated systems for generating building forms—transforming single 2D images into EnergyPlus models.

Contribution: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.

HONORS & AWARDS

National Inspirational Scholarship	2022/12
Shanghai Jiao Tong University Scholarship	2022/12
National Inspirational Scholarship	2021/12
Shanghai Jiao Tong University Scholarship	2021/12
Third Prize in Shanghai Jiao Tong University Structure Competition	2020/12
Second Prize in the 2020 Li Zhengdao Science and Art Competition	2020/11
Third Prize in Shanghai Jiao Tong University Structure Competition	2019/11
First Prize in the 2019 Li Zhengdao Science and Art Competition	2019/11
Second Prize in Tongji University Wooden Structure Competition	2019/05

PROFESSIONAL SERVICE

External reviewer	Journal of Building Engineering (since 2022)
External reviewer	Building and Environment (since 2025)

PROGRAM LANGUAGES OR SOFTWARE SKILL

- Python
- Grasshopper
- EnergyPlus
- CFD

LANGUAGES

- English
- Chinese